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The Flash Crash: High-Frequency Trading in an Electronic Market

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1 Big picture

- **Question.** How did nondesignated intraday intermediaries behave before and during the Flash Crash, a period of large and temporary selling pressure in the E-mini S&P 500 futures market?
- **Setting.** The paper studies the E-mini futures market around May 6, 2010 using account-level audit-trail transaction data for May 3–6. The focus is on the triggering venue emphasized in the joint CFTC-SEC narrative of the Flash Crash.
- **Core challenge.** In an automated limit-order market with no formal market-maker obligations, one cannot simply label traders as "market makers" and infer their role from designation. The paper has to classify traders from behavior, then distinguish between traditional inventory-based intermediation and fast-trading strategies that trade ahead of short-run price changes.
- **Main theoretical comparison.** The paper evaluates the Flash Crash through two lenses:
  - *limited risk-bearing capacity* of intermediaries during a liquidity shock, and
  - *stale-quote sniping / latency arbitrage*, in which faster traders pick off stale quotes and therefore trade in the direction of imminent price changes.
- **Main empirical strategy.** First, classify traders into High-Frequency Traders, Market Makers, Fundamental Buyers, Fundamental Sellers, Opportunistic Traders, and Small Traders using inventory and volume patterns from May 3–5. Second, keep those labels fixed on May 6 and study second-by-second comovement between inventory changes and price changes. Third, examine event-time price responses after HFT trades and trader participation around discrete price changes.
- **Main finding.** Intraday intermediaries had limited inventory capacity relative to the 75,000-contract sell program. The most active intermediaries, classified as High-Frequency Traders (HFTs), did *not* significantly change their inventory dynamics during the price decline itself. In contrast, Market Makers' inventory-price relationship did change during the Flash Crash.
- **Key cross-sectional finding.** Market Makers look like textbook liquidity suppliers: their inventory changes are negatively related to contemporaneous price changes and positively related to subsequent reversals. HFT inventory changes look different: they are positively related to contemporaneous prices and short-horizon future prices.
- **Main microstructure result.** When HFTs are net buyers in a second, prices rise in the following second and remain elevated for up to 20 seconds. Around discrete price changes, HFTs disproportionately trade aggressively against the old quote and then provide a disproportionate share of passive depth at the new quote.
- **Why this matters.** The paper argues that the most active intraday intermediaries during the Flash Crash were not stabilizing the market in the way a textbook dealer would. Instead, they appear more consistent with fast traders exploiting stale quotes while carrying limited inventory.
- **Economic takeaway.** The Flash Crash is not evidence that HFTs suddenly "turned off." The paper's stronger message is that HFTs and traditional Market Makers are different animals: the former kept trading in roughly the same short-horizon style, while the latter altered their inventory-price dynamics when stress intensified.

2 Data and measurement (key objects)

2.1 Institutional background: the E-mini market

- **Asset.** June 2010 E-mini S&P 500 futures contract.
- **Venue.** CME Globex, a fully electronic centralized limit-order market.
- **Contract size.** One E-mini contract is worth $50 times the S&P 500 index. Around May 3–6, 2010, with the index slightly above 1,000, each contract was worth about $50,000.
- **Tick size.** 0.25 index points, that is, $12.50 per contract.
- **Matching rule.** Price-time priority.
- **Transparency.** The market has both pretrade and posttrade transparency, but trader identities are not publicly disclosed.
- **Why this market is ideal for the paper.**
  - All trading takes place in *one venue*, so the authors avoid fragmented-market measurement problems.
  - There are *no formal market-maker obligations*, so liquidity supply must be inferred from actual behavior rather than designation.
  - The E-mini is the main price-discovery venue among U.S. equity-index products.

2.2 The Flash Crash episode itself

- At 2:32 p.m. CT on May 6, 2010, a large mutual-fund complex initiated an automated program to sell 75,000 E-mini contracts, roughly $4.1 billion notional, without regard to price or time.
- Between 2:32 p.m. and 2:45 p.m., the algorithm sold about 35,000 contracts.
- By 2:45:28 p.m., less than 1,050 contracts remained as buy-side resting orders in the E-mini, under 1% of the buy-side depth seen at the beginning of the day.
- At 2:45:28 p.m., CME Stop Logic paused trading for five seconds; prices stabilized after the reopening and then recovered.

Interpretation. The paper treats this as a clean episode of large and temporary selling pressure in the central futures market where the systemic event began.

2.3 Sample and raw data

- **Sample period.** May 3 through May 6, 2010.
- **Intraday window each day.** 8:30 a.m. CT to 3:15 p.m. CT, a 405-minute window beginning with the opening of the underlying stock market and ending at the CME technical maintenance break.
- **Source.** CME Trade Capture Report (TCR) provided to the CFTC.
- **Unit of observation.** Executed transaction.

- **Fields used.** Buyer and seller account identifiers, price, quantity, second timestamp, within-second sequence ID, market/limit indicator, order ID linking partial executions to the original order, and a passive/aggressive flag from Globex.
- **Crucial limitation.** The authors do *not* observe nonexecuted messages. So they do not see the full limit-order-book message flow, only executions.

2.4 Market descriptive statistics

- Average daily volume on May 3–5 was 2,397,639 contracts; on May 6 it was 5,094,703.
- The number of trades rose from 446,340 to 1,030,204.
- The number of active trading accounts rose from 11,875 to 15,422.
- Average trade size was about five contracts on both sides of the event (5.41 before, 4.99 on May 6).
- More than 90% of both trades and volume were executed via limit orders on all days.
- Intraday volatility, measured as the log high-low price range, jumped from 1.54% on May 3–5 to 9.82% on May 6.

Interpretation. The Flash Crash day is not just a big-price-drop day; it is also a day of dramatically larger trading intensity and much higher volatility, but still within an overwhelmingly limit-order-driven market structure.

2.5 Trader classification

The paper classifies more than 15,000 accounts using behavior on May 3–5 and then keeps the categories fixed on May 6.

Step 1: identify intraday intermediaries. A trader is an intraday intermediary if it:

1. trades at least 10 contracts on at least one of May 3, 4, or 5,

$$Vol_d \geq 10, \tag{1}$$

2. has a small end-of-day directional position relative to daily trading volume,

$$\frac{1}{3}\sum_{d=1}^{3}\left|\frac{NP_{d,405}}{Vol_d}\right| \leq 5\%, \tag{2}$$

3. and exhibits intraday mean reversion in inventories,

$$\frac{1}{3}\sum_{d=1}^{3}\sqrt{\frac{1}{405}\sum_{t=1}^{405}\left(\frac{NP_{d,t}-NP_{d,405}}{Vol_d}\right)^2} \leq 0.5\%. \tag{3}$$

Here $NP_{d,t}$ is the trader's net position during day d at minute t, and Vol_d is the day's total trading volume for the account.

Step 2: split intermediaries into HFTs and Market Makers.

- The **16 most active intraday intermediaries** by number of trades over May 3–5 are called **High-Frequency Traders**.
- The remaining intraday intermediaries are called **Market Makers**.

Step 3: classify everyone else.

- **Small Traders:** fewer than 10 contracts on that day.
- **Fundamental Buyers:** trade at least 10 contracts and finish with a long end-of-day position of at least 15% of daily volume.
- **Fundamental Sellers:** trade at least 10 contracts and finish with a short end-of-day position of at least 15% of daily volume.
- **Opportunistic Traders:** residual category.

Interpretation.

- The key design choice is that the classification is based on *pre-crash* behavior. This prevents the authors from classifying traders by the very behavior they later want to explain.
- A High-Frequency Trader is not just "a fast trader" in the colloquial sense. It is a trader that looks like an intraday intermediary but participates in a markedly larger fraction of trading than ordinary Market Makers.

2.6 What the categories look like in the data

- Over May 3–5, HFTs account for 34.22% of total trading volume and 32.56% of trades, while Market Makers account for another 10.49% of volume and 11.63% of trades.
- On May 6, those shares are 28.57% and 29.35% for HFTs, and 9.00% and 11.48% for Market Makers.
- HFTs are much more aggressive than Market Makers: the volume-weighted aggressive share is 49.86% for HFTs versus 34.99% for Market Makers over May 3–5, and 46.59% versus 32.49% on May 6.
- HFTs are very small in count but very large in activity: about 15 HFT accounts versus about 189 Market Makers pre-crash.
- HFTs trade far more frequently: about 5.98 trades per second on average versus 2.14 for Market Makers, implying roughly a 30-fold difference in frequency at the account level.

Interpretation. The HFT category is not just a slightly more active subset of Market Makers. It is a tiny but exceptionally active set of accounts with very high participation and a much more aggressive execution profile.

3 Models and empirical specifications (all equations + interpretation)

3.1 This is an empirical paper, not a full structural model

The paper is driven by theory, but it does not estimate a structural equilibrium model of the Flash Crash. Instead, its "model" section is really a set of theoretically motivated empirical designs.

The two main theoretical benchmarks are:

- **Limited risk-bearing capacity.** If a sudden order imbalance overwhelms intermediaries, prices must move enough to induce reluctant intermediaries to absorb inventory.
- **Stale-quote sniping / latency arbitrage.** If some traders react faster than others, they may trade in the same direction as near-future price changes by picking off stale quotes.

Interpretation. The paper's contribution is to ask which of these theoretical patterns better describes the behavior of different classes of traders during the Flash Crash.

3.2 Baseline inventory-price regression

The main second-by-second regression for category-specific inventories is

$$\Delta y_t = a + \phi\, \Delta y_{t-1} + \delta\, y_{t-1} + \sum_{i=0}^{20} \xi_i \left(\frac{\Delta p_{t-i}}{0.25} \right) + e_t, \tag{1}$$

where:

- y_t is the net inventory of HFTs or Market Makers at second t,
- Δy_t is the second-by-second inventory change,
- Δp_t is the change in the high-low midpoint from second $t-1$ to second t,
- dividing by 0.25 converts price changes into *ticks*.

Interpretation of coefficients.

- $\delta < 0$ means inventories are mean reverting.
- A negative contemporaneous ξ_0 means traders buy when prices are temporarily low and sell when prices are high, the classic dealer pattern.
- A positive contemporaneous ξ_0 means inventory changes move *with* prices, which is more consistent with trading ahead of price changes.
- The lag profile tells us whether a category's inventory response is mostly contemporaneous, quickly reversing, or persistent over short horizons.

What Table III says. For May 3–5, the estimated contemporaneous coefficient on price changes is:

- **HFTs:** +32.09,

- **Market Makers:** -13.54.

For HFTs, the next few lags are also positive (17.18, 8.36, 5.09, 3.91), and later lags turn negative. For Market Makers, the contemporaneous coefficient is negative and the next few lags are positive (-1.22, 2.16, 2.53, 2.65, 2.50, ...).

Interpretation.

- HFT inventories positively comove with current and near-future prices.
- Market Maker inventories negatively comove with current prices but positively with later price changes, which is exactly the pattern of absorbing order flow and then unwinding as prices partially reverse.

3.3 Flash-Crash interaction regression

To test whether the inventory-price relationship changed during the Flash Crash, the authors estimate

$$\Delta y_t = a + \phi\,\Delta y_{t-1} + \delta\, y_{t-1} + \sum_{i=0}^{20} \xi_i \left(\frac{\Delta p_{t-i}}{0.25}\right) \tag{4}$$

$$+ D_t^D \left[a^D + \phi^D \Delta y_{t-1} + \delta^D y_{t-1} + \sum_{i=0}^{20} \xi_i^D \left(\frac{\Delta p_{t-i}}{0.25}\right)\right] \tag{5}$$

$$+ D_t^U \left[a^U + \phi^U \Delta y_{t-1} + \delta^U y_{t-1} + \sum_{i=0}^{20} \xi_i^U \left(\frac{\Delta p_{t-i}}{0.25}\right)\right] + e_t, \tag{6}$$

where D_t^D marks the *down* phase of the Flash Crash (13:32:00–13:45:28 CT) and D_t^U marks the *up* phase after the five-second trading pause (13:45:33–14:08:00 CT).

Interpretation.

- The interaction coefficients ask whether the HFT or Market Maker inventory-price relation shifted relative to normal days.
- For HFTs, the key result is that the *down-phase interactions are almost all insignificant.* In other words, when prices were actually collapsing, HFT trading dynamics looked much like on normal days.
- For Market Makers, the interaction coefficients are significant in both the down and up phases; the lag structure shifts during the crash.

Main inference from Table IV.

- For HFTs, the authors *fail to reject* that the interaction coefficients are jointly zero.
- For Market Makers, they *reject* joint equality at the 1% level.

Economic meaning. The paper's strongest distinction is here: the crash changed Market Maker dynamics, not HFT dynamics.

3.4 Inventory bounds as evidence on risk-bearing capacity

This part is not a regression but a quantitative benchmark.

- Combined net inventories of all classified intraday intermediaries over the whole four-day sample never exceeded about 6,000 contracts.
- HFTs never accumulated inventories larger than roughly 4,000 contracts.
- Market Makers never exceeded about 1,500 contracts in either direction.
- These numbers are tiny relative to the 75,000-contract sell program and even small relative to the roughly 30,000-contract imbalance between Fundamental Sellers and Fundamental Buyers documented around the crash window.

Interpretation. This is the paper's most direct evidence for limited intermediary risk-bearing capacity. Intermediaries simply did not have the balance-sheet capacity to absorb the entire liquidity shock.

3.5 Event-time price response after HFT trades

The next design studies how prices move *after* HFTs trade. For an event-second in which HFTs are net buyers, net aggressive buyers, or net passive buyers, the paper computes the difference between:

- the value-weighted average price paid by HFTs in that second, and
- the value-weighted average price for all trades in the same second and each of the following 20 seconds.

Results are weighted by the magnitude of the HFT net position change in the event second.

Interpretation. If prices systematically move in the same direction *after* HFTs trade, then HFTs are trading ahead of short-run price changes rather than just mechanically moving the price on impact.

Figure 4 results.

- When HFTs are net buyers over May 3–5, prices rise by about 17% of a tick in the next second and remain about 15% of a tick higher even 20 seconds later.
- For **aggressive** HFT buys, prices rise by about 20% of a tick in the next second, rise further to roughly 23% of a tick over seconds 2–11, and remain about 15% of a tick higher after 20 seconds.
- For **passive** HFT buys, prices rise only about 2% of a tick in the next second and then drift down to roughly −3% of a tick by second 20.
- For HFT sell events, the pattern is roughly symmetric. The striking exception is passive HFT sells: prices fall at first but then drift *upward*, reaching about 12% of a tick above the event-second price after 19 seconds.

Interpretation. Aggressive HFT trades line up with upcoming price movements, while passive HFT trades do not. This is exactly what one would expect if HFTs are taking stale quotes and then recycling inventory at the new price level.

3.6 Price-change-event participation design

To test stale-quote sniping more directly, the paper defines:

- a **price increase event** when the best ask rises, and
- a **price decrease event** when the best bid falls.

For each event, the authors:

1. count backward the **last 100 contracts** traded at the old quote before the price change,
2. count forward the **first 100 contracts** traded at the new quote after the price change,
3. decompose both sides of those trades into aggressive and passive participation by each trader category.

Interpretation. A quote-sniping trader should trade aggressively against the old quote just before the quote moves and then appear disproportionately on the passive side at the new quote.

3.7 What this section as a whole delivers

- The regressions deliver evidence on *how inventories covary with prices.*
- The inventory bounds deliver evidence on *how much risk intermediaries can bear.*
- The event-time and quote-change designs deliver evidence on *how HFTs actually trade at very short horizons.*

Taken together, they separate three ideas that are often blurred together in policy discussion: "providing liquidity," "bearing inventory risk," and "trading very fast."

4 Identification and instruments (what makes the estimates credible)

4.1 There is no IV here – identification comes from design

This paper is not an instrumental-variables paper. Credibility comes from institutional setting, account-level classification, and comparison of the same traders before and during the crash.

4.2 Why the E-mini setting is unusually clean

- **Single centralized venue.** The E-mini trades in one electronic market, so the paper sees the complete universe of executions in that contract.
- **No market-maker obligations.** This matters because any liquidity-supply behavior is voluntary, not mechanically induced by formal obligations.
- **Account-level IDs.** The authors do not infer HFT behavior from partial exchange-specific flags or from aggregate trade patterns. They observe the entire account-level execution history in the contract.

4.3 Why the classification is credible

- Classification is based on *May 3–5* inventory and volume behavior, not May 6 crash behavior.
- The definitions do not use price comovement or the Flash Crash itself; they use only trading intensity and inventory patterns.
- Keeping categories fixed on May 6 prevents an obvious endogeneity problem in which traders would be labeled by the outcome the paper is trying to explain.

4.4 What makes the main estimates persuasive

- The baseline and interaction regressions compare the same trader types on normal days and crash days at one-second frequency.
- The event-study result is stronger than a contemporaneous correlation because prices move *after* HFT trades, not just during the event second.
- The quote-change design is especially compelling because it ties HFT behavior to a concrete microstructure pattern: taking the old quote, then providing at the new quote.

4.5 What the results do and do not identify

- The baseline inventory-price regressions study *comovement*, not causality.
- Without message-level data, the paper cannot prove whether HFTs react to public information, private signals, or other traders' order revisions.
- The HFT category is likely a *subset* of all possible high-frequency trading strategies.
- The evidence is strongest on what HFTs were *not*: they were not acting like classic stabilizing inventory dealers during the price collapse.

4.6 Main limitations

- No message data: one cannot directly observe quote updates, cancellations, or latency.
- No cross-market position data for every account: part of the imbalance may be absorbed by traders doing cross-market arbitrage, especially Opportunistic Traders.
- The paper studies one dramatic event in one contract. It is highly informative, but external validity must be judged carefully.

4.7 Relation to the literature

- Relative to the *traditional market-making* literature, the paper shows that not all active liquidity suppliers in electronic markets behave like classical dealers.
- Relative to the *Flash Crash* literature, the paper focuses on the triggering venue and on the behavior of intermediary categories rather than on cross-market propagation or broken trades.
- Relative to the *HFT* literature, the paper's contribution is to connect account-level HFT trade patterns to theories of quote sniping during a systemic liquidity event.

5 Tables and figures

5.1 Figure 1 and Table I: what made May 6 different

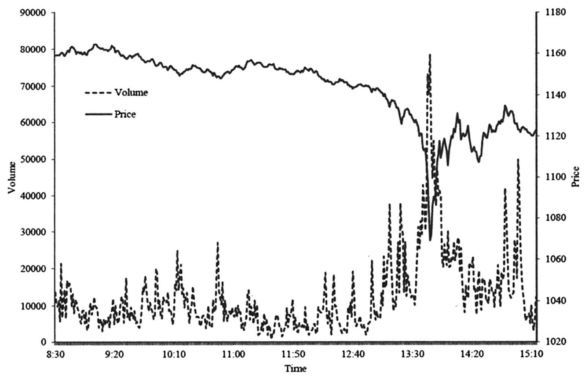

Figure 1. Prices and trading volume of the E-mini S&P 500 stock index futures contract. This figure presents minute-by-minute transaction prices and trading volume of the June 2010 E-mini S&P futures contract on May 6, 2010, between 8:30 and 15:15 CT. Trading volume is calculated as the number of contracts traded during each minute. Transaction price is the last transaction price of each minute. Source: "Preliminary Findings Regarding the Market Events of May 6, 2010."

Table I
Market Descriptive Statistics

This table presents summary statistics for the June 2010 E-mini S&P 500 futures contract. The first column presents averages calculated for May 3 through 5, 2010, between 8:30 and 15:15 CT. The second column presents statistics for May 6, 2010, between 8:30 and 15:15 CT. Volume is the number of contracts traded. The number of traders is the number of trading accounts that traded at least once during a trading day. Order size and trade size are measured in number of contracts. The use of limit orders is presented in both percent of the number of transactions and trading volume. Volatility is calculated as the natural logarithm of maximum price over minimum price within a trading day.

	May 3–5	May 6
Daily trading volume	2,397,639	5,094,703
# of trades	446,340	1,030,204
# of traders	11,875	15,422
Trade size	5.41	4.99
Limit orders % volume	95.45%	92.44%
Limit orders % trades	94.36%	91.75%
Volatility (log high-low price range)	1.54%	9.82%
Return	−0.02%	−3.05%

Read:

- Figure 1 shows the crash itself in the E-mini: a sharp drop in price and a huge spike in trading volume.
- Table I quantifies how abnormal the day was. Daily volume more than doubled from 2.40 million to 5.09 million contracts, and the number of trades rose from 446,340 to over 1.03 million.
- Volatility jumped from 1.54% to 9.82%.
- The average trade size barely moved, remaining around five contracts.
- More than 90% of activity still occurred through limit orders.

Why it matters. The event is not driven by a disappearance of ordinary exchange mechanics. The market stayed electronically active and overwhelmingly limit-order based; what changed was the scale of order imbalance and price movement.

5.2 Figure 2: the ecosystem of traders

Read:

- Figure 2 plots accounts by trading volume and end-of-day net position scaled by market trading volume for each of the four days.
- Five clusters emerge: High-Frequency Traders, Market Makers, Fundamental Buyers, Fundamental Sellers, and Opportunistic Traders.
- HFTs stand out sharply in the vertical direction because their trading activity is much larger than that of all other categories.
- Fundamental Sellers spread to the left because they accumulate large negative end-of-day positions, while Fundamental Buyers spread to the right.

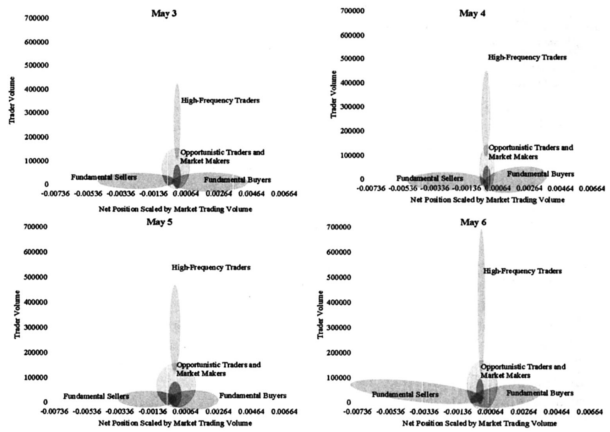

Figure 2. Trading accounts, trading volume, and net position scaled by market trading volume. This figure presents trader categories superimposed (as shaded areas) over all individual trading accounts ranked by their trading volume and net position scaled by market trading volume. The panels reflect trading activity in the June 2010 E-mini S&P 500 futures contract for May 3 through 6, 2010.

- Opportunistic Traders overlap with multiple groups, which fits the idea that this category mixes several strategies.

Why it matters. The classification is not just a theoretical labeling exercise. The categories appear as distinct clusters in the data, especially HFTs versus Market Makers.

5.3 Table II: summary statistics of trader categories

Table II
Summary Statistics of Trader Categories

This table presents summary statistics for trader categories. Panel A presents three-day average statistics for May 3 through 5, 2010, from 8:30 to 15:15 CT. Percentage of trading volume is a three-day average of the daily percentage of total trading volume for each trader category. Percentage of trades is a three-day average of the daily percentage of total trades for each trader category. Trade Size (Avg.) is a three-day average of the daily account-level average trade size within each trader category. Order Size (Avg.) is a three-day average of the daily account-level average size of the executed portion of an order within each trader category. Limit Orders % of Volume is a three-day average of the percentage of trader category trading volume that resulted from marketable and nonmarketable limit orders. Agg Ratio Vol-Weighted is a three-day average of the percentage of trader category trading volume that resulted from marketable orders. Panel B presents statistics for May 6 from 8:30 to 15:15 CT.

Panel A: May 3–5 Three-Day Average

Trader Type	% Volume	% of Trades	# Traders	Trade Size (Avg.)	Order Size (Avg.)	Limit Orders % Volume	Agg Ratio Vol-Weighted
High-frequency traders	34.22%	32.56%	15	5.69	14.75	100.00%	49.86%
Market makers	10.49%	11.63%	189	4.88	7.92	99.61%	34.99%
Fundamental buyers	11.89%	10.15%	1,013	6.34	14.09	91.26%	58.40%
Fundamental sellers	12.11%	10.10%	1,088	6.50	14.20	92.18%	54.98%
Opportunistic traders	30.79%	33.34%	3,504	4.98	8.80	92.14%	50.49%
Small traders	0.50%	2.22%	6,065	1.22	1.25	70.09%	58.54%
	Volume	# of Trades	# Traders	Trade Size (Avg.)	Order Size (Avg.)	Limit Orders % Volume	
All	2,397,639	446,340	11,875	5.41	10.83	95.45%	

(*Continued*)

Table II—*Continued*

Panel B: May 6

Trader type	% Volume	% of Trades	# Traders	Trade Size (Avg.)	Order Size (Avg.)	Limit Orders % Volume	Agg Ratio Vol-Weighted
High-frequency traders	28.57%	29.35%	16	4.85	9.86	100.00%	46.59%
Market makers	9.00%	11.48%	179	3.89	5.88	99.64%	32.49%
Fundamental buyers	12.01%	11.54%	1,263	5.15	10.43	88.84%	56.43%
Fundamental sellers	10.04%	6.95%	1,276	7.19	21.29	89.99%	55.30%
Opportunistic traders	40.13%	39.64%	5,808	5.05	10.06	87.39%	52.98%
Small traders	0.25%	1.04%	6,880	1.20	1.24	63.61%	63.63%
	Volume	# of Trades	# Traders	Trade Size (Avg.)	Order Size (Avg.)	Limit Orders % Volume	
All	5,094,703	1,030,204	15,422	4.99	9.76	92.443%	

Read:

- HFTs are only about 15 accounts pre-crash, but they account for 34.22% of volume and 32.56% of trades.
- Market Makers are far more numerous (about 189 accounts) but account for only 10.49% of volume.
- On May 6, HFT share falls somewhat to 28.57% of volume and Market Maker share to 9.00%.
- HFTs are materially more aggressive than Market Makers: 49.86% versus 34.99% aggressive volume pre-crash, and 46.59% versus 32.49% on May 6.
- HFT and Market Maker activity is overwhelmingly limit-order based, but HFTs are much closer to a 50-50 aggressive/passive split than Market Makers.

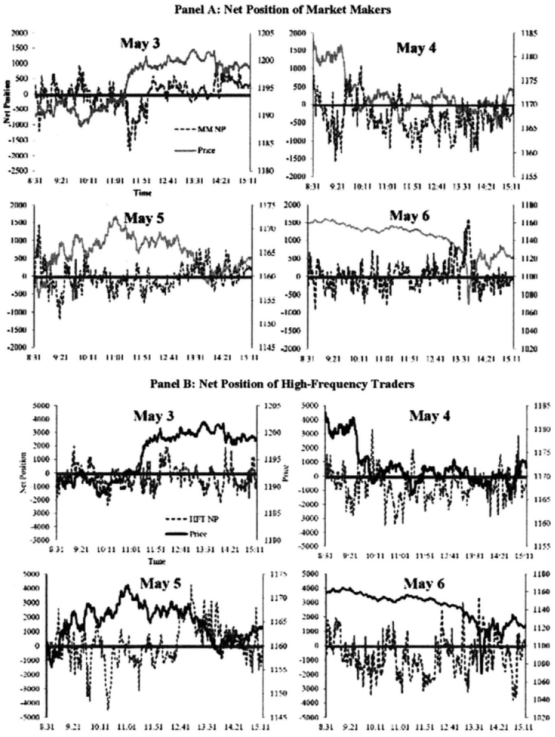

Figure 3. Net position of Market Makers and High-Frequency Traders. This figure presents the net position (left vertical axis) of Market Makers and High-Frequency Traders and market transaction prices (right vertical axis) in the June 2010 E-mini S&P 500 futures contract over one-minute intervals during May 3, 4, 5, and 6 between 8:30 and 15:15 CT. Net position is calculated as the difference between the total open long and total open short positions of Market Makers and High-Frequency Traders at the end of each minute. Transaction price is the last market transaction price each minute. The top panel presents the net positions of Market Makers and the bottom panel presents the net positions of High-Frequency Traders.

Why it matters. This table is the first warning against treating HFTs and Market Makers as one homogeneous liquidity-supplying class.

5.4 Figure 3: inventory bounds and risk-bearing capacity

Read:

- Figure 3 plots end-of-minute net positions for Market Makers and HFTs together with prices over May 3–6.
- HFT inventories remain bounded within roughly ±4,000 contracts even on the Flash Crash day.
- Market Maker inventories remain even tighter, within roughly ±1,500 contracts.
- The figure shows that neither group scaled up risk taking enough to absorb the full sell pressure.
- A key detail is that shortly before Stop Logic, HFTs liquidated about 2,000 contracts they had accumulated earlier, while Market Makers did not liquidate until after the pause.

Why it matters. Figure 3 is the paper's visual evidence for the limited-inventory-capacity story. Intermediaries were active, but their balance sheets were simply small relative to the shock.

Table III

Baseline Regression: Net Holdings and Prices

This table presents estimated coefficients for the regression: $\Delta y_t = \alpha + \phi \Delta y_{t-1} + \delta y_{t-1} + \sum_{i=0}^{20} [\beta_i \times \Delta p_{t-i}/0.25] + \epsilon_t$. The dependent variable is the change in holdings of High-Frequency Traders or Market Makers, as indicated. Both changes in holdings, Δy_t, and lagged holdings, y_{t-1}, are in contracts. Price changes, Δp_{t-i}, are in ticks. The sampling frequency is one second. t-statistics, calculated using the White (1980) estimator are reported in parentheses. Observations are stacked for May 3 through 5.

	Δ *NP HFT*	Δ *NP MM*
Intercept	−1.64	−0.53
	(−3.54)	(−3.33)
$\Delta NPHFT_{t-1}$	−0.01	
	(−0.69)	
$NPHFT_{t-1}$	−0.01	
	(−11.77)	
$\Delta NPMM_{t-1}$		
		(−0.79)
$NPMM_{t-1}$		−0.004
		(−8.93)
ΔP_t	32.09	−13.54
	(18.44)	(−23.83)
ΔP_{t-1}	17.18	−1.22
	(12.58)	(−2.71)
ΔP_{t-2}	8.36	2.16
	(7.15)	(4.99)
ΔP_{t-3}	5.09	2.53
	(4.93)	(5.97)
ΔP_{t-4}	3.91	2.65
	(3.62)	(6.54)
ΔP_{t-5}	1.81	2.50
	(1.56)	(5.91)
ΔP_{t-6}	−0.08	2.16
	(−0.07)	(5.42)
ΔP_{t-7}	−1.00	1.84
	(−0.97)	(4.96)
ΔP_{t-8}	−1.76	1.47
	(−1.56)	(3.83)
ΔP_{t-9}	−1.81	0.45
	(−1.70)	(1.19)
ΔP_{t-10}	−3.90	0.52
	(−3.78)	(1.37)
ΔP_{t-11}	−4.73	−0.03
	(−4.70)	(−0.07)
ΔP_{t-12}	−3.46	0.15
	(−3.33)	(0.41)
ΔP_{t-13}	−3.80	0.27
	(−3.74)	(0.72)
ΔP_{t-14}	−4.77	0.32
	(−4.70)	(0.86)
ΔP_{t-15}	−2.74	−0.19
	(−2.63)	(−0.53)

(Continued)

Table III—*Continued*

	Δ *NP HFT*	Δ *NP MM*
ΔP_{t-16}	−2.21	−0.64
	(−2.09)	(−1.72)
ΔP_{t-17}	−2.52	−0.10
	(−2.45)	(−0.26)
ΔP_{t-18}	−4.36	0.04
	(−3.96)	(0.12)
ΔP_{t-19}	−4.21	0.57
	(−4.16)	(1.51)
ΔP_{t-20}	−5.86	−0.12
	(−5.86)	(−0.33)
#obs	72,837	72,837
Adj-R^2	0.019	0.026

5.5 Table III and Table IV: who looked like a textbook market maker?

Read:

- Table III shows that Market Makers have the classic sign pattern: inventory changes are negatively related to contemporaneous prices and positively related to subsequent price changes.
- HFTs look very different: their contemporaneous coefficient is strongly positive (32.09), and the first few lagged coefficients are also positive before turning negative later.
- Table IV then asks whether these patterns changed during the Flash Crash itself.
- For HFTs, the answer is mostly no during the *down* phase. The joint change is not statistically compelling.
- For Market Makers, the answer is yes. Their lag structure shifts significantly in both down and up phases, and the joint null of no change is rejected at the 1% level.

Table IV

High-Frequency Traders and Market Makers: The Flash Crash

This table presents estimated coefficients for the regression: $\Delta y_t = \alpha + \phi\Delta y_{t-1} + \Delta\gamma_{t-1} + \Delta\gamma_{t-1} + \Sigma_{i=0}^{20}[\delta_i \times p_{t-i}/0.25] + D_t^D|\alpha^D + \phi^D\Delta y_{t-1} + \delta^D y_{t-1} + \Sigma_{i=0}^{20}[\delta_i^D \times p_{t-i}/0.25)] + D_t^U|\alpha^U + \phi^U\Delta y_{t-1} + \delta^U y_{t-1} + \Sigma_{i=0}^{20}[\delta_i^U \times p_{t-i}/0.25)] + \epsilon_t$ over May 3 through 6, 2010, with dummy variables D_t^D and D_t^U included to interact with observations during the "Down" (from 13:32:00 CT to 13:45:28 CT) and "Up" (from 13:45:33 CT to 14:08:00 CT) phases of the Flash Crash. The period between 13:45:28 CT and 13:45:33 CT corresponds to the five-second pause in trading; there are no changes in prices or inventory during the five-second pause. The cutoff for observations on May 6, 2010, is 14:08:00 CT. The dependent variable is the change in holdings of High-Frequency Traders or Market Makers, as indicated. Both changes in holdings, Δy_t, and lagged holdings, y_{t-1}, are in contracts. Price changes, Δp_{t-i}, are in ticks. Estimates are computed for second-by-second observations. t-statistics, calculated using the White (1980) estimator are reported in parentheses.

Variable	Δ *NP HFT*	Δ *NP MM*	Variable (cont)	Δ *NP HFT*	Δ *NP MM*	Variable (cont)	Δ *NP HFT*	Δ *NP MM*
Intercept	−2.04	−0.48	Intercept^D	9.22	9.15	Intercept^U	2.27	0.49
	(−4.78)	(−3.34)		(1.19)	(2.41)		(0.55)	(0.33)
ΔNP_{t-1}	−0.005	−0.024	ΔNP_{t-1}^D	−0.031	−0.024	ΔNP_{t-1}^U	0.004	0.085
	(−0.69)	(−3.31)		(−0.80)	(−0.63)		(0.10)	(2.74)
NP_{t-1}	−0.005	−0.005	NP_{t-1}^D	−0.002	−0.007	NP_{t-1}^U	−0.001	0.000
	(−12.95)	(−10.78)		(−0.38)	(−1.62)		(−0.21)	(−0.17)
ΔP_t	31.47	−15.48	ΔP_t^D	1.29	14.13	ΔP_t^U	−40.83	14.29
	(16.89)	(−21.96)		(0.18)	(6.73)		(−12.18)	(13.68)
ΔP_{t-1}	14.96	−0.54	ΔP_{t-1}^D	−3.02	11.44	ΔP_{t-1}^U	−9.60	5.63
	(12.17)	(−1.23)		(−0.57)	(5.11)		(−3.44)	(7.12)
ΔP_{t-2}	6.24	2.69	ΔP_{t-2}^D	−6.84	1.87	ΔP_{t-2}^U	−9.72	−1.83
	(5.36)	(5.99)		(−1.26)	(0.81)		(−3.57)	(−2.20)
ΔP_{t-3}	3.02	2.65	ΔP_{t-3}^D	−4.16	−2.03	ΔP_{t-3}^U	−3.97	−2.47
	(3.31)	(7.14)		(−0.69)	(−1.22)		(−1.61)	(−3.75)
ΔP_{t-4}	1.92	2.74	ΔP_{t-4}^D	−9.74	−4.91	ΔP_{t-4}^U	−1.12	−2.51
	(2.04)	(7.78)		(−1.98)	(−3.11)		(−0.49)	(−3.70)
ΔP_{t-5}	0.63	2.21	ΔP_{t-5}^D	−10.94	−3.45	ΔP_{t-5}^U	1.86	−2.86
	(0.64)	(5.99)		(−1.57)	(−2.25)		(0.75)	(−4.36)
ΔP_{t-6}	−1.89	1.99	ΔP_{t-6}^D	0.59	−2.91	ΔP_{t-6}^U	4.27	−2.45
	(−2.03)	(5.72)		(0.11)	(−1.86)		(1.78)	(−3.71)

(*Continued*)

Table IV—*Continued*

Variable	Δ *NP HFT*	Δ *NP MM*	Variable (cont)	Δ *NP HFT*	Δ *NP MM*	Variable (cont)	Δ *NP HFT*	Δ *NP MM*
ΔP_{t-7}	−2.85	1.92	ΔP_{t-7}^D	−1.66	−2.71	ΔP_{t-7}^U	−4.54	−3.38
	(−2.89)	(5.18)		(−0.31)	(−1.59)		(−1.73)	(−5.05)
ΔP_{t-8}	−2.52	1.43	ΔP_{t-8}^D	2.45	−2.97	ΔP_{t-8}^U	1.79	−1.65
	(−2.68)	(4.33)		(0.44)	(−1.92)		(0.76)	(−2.76)
ΔP_{t-9}	−2.59	0.48	ΔP_{t-9}^D	−4.32	−2.98	ΔP_{t-9}^U	2.69	−1.64
	(−2.76)	(1.44)		(−0.61)	(−1.70)		(1.12)	(−2.54)
ΔP_{t-10}	−5.18	0.91	ΔP_{t-10}^D	3.93	−3.40	ΔP_{t-10}^U	4.41	−1.52
	(−4.66)	(2.12)		(0.50)	(−1.78)		(1.76)	(−2.22)
ΔP_{t-11}	−5.07	−0.05	ΔP_{t-11}^D	9.84	−6.35	ΔP_{t-11}^U	6.01	−0.36
	(−5.76)	(−0.16)		(1.30)	(−2.96)		(2.27)	(−0.51)
ΔP_{t-12}	−4.05	−0.10	ΔP_{t-12}^D	8.38	−0.73	ΔP_{t-12}^U	4.37	−0.79
	(−4.46)	(−0.31)		(1.07)	(−0.37)		(1.34)	(−1.26)
ΔP_{t-13}	−3.86	−0.07	ΔP_{t-13}^D	11.92	−4.69	ΔP_{t-13}^U	10.02	0.28
	(−4.27)	(−0.20)		(1.64)	(−2.10)		(3.34)	(0.43)
ΔP_{t-14}	−4.36	0.28	ΔP_{t-14}^D	−8.56	0.79	ΔP_{t-14}^U	1.64	−0.59
	(−5.01)	(0.84)		(−1.29)	(0.41)		(0.62)	(−0.98)
ΔP_{t-15}	−2.05	−0.17	ΔP_{t-15}^D	8.46	−5.41	ΔP_{t-15}^U	1.47	−0.09
	(−2.27)	(−0.50)		(1.17)	(−2.55)		(0.64)	(−0.15)
ΔP_{t-16}	−2.01	−0.39	ΔP_{t-16}^D	−3.25	3.92	ΔP_{t-16}^U	1.07	0.99
	(−2.10)	(−1.11)		(−0.41)	(1.80)		(0.37)	(1.56)
ΔP_{t-17}	−2.67	0.01	ΔP_{t-17}^D	6.24	−1.57	ΔP_{t-17}^U	5.19	0.48
	(−3.05)	(0.02)		(0.81)	(−0.69)		(2.13)	(0.75)
ΔP_{t-18}	−3.89	0.19	ΔP_{t-18}^D	−8.62	0.86	ΔP_{t-18}^U	7.37	−0.69
	(−4.10)	(0.58)		(−1.05)	(0.42)		(2.58)	(−1.12)
ΔP_{t-19}	−3.50	0.70	ΔP_{t-19}^D	−1.05	−3.07	ΔP_{t-19}^U	−0.75	−0.88
	(−3.88)	(2.08)		(−0.12)	(−1.39)		(−0.30)	(−1.44)
ΔP_{t-20}	−5.30	−0.33	ΔP_{t-20}^D	−2.32	3.13	ΔP_{t-20}^U	4.88	−0.06
	(−5.82)	(−1.00)		(−0.30)	(1.36)		(2.14)	(−0.09)
			# of obs	93,092	93,092			
			Adj-R^2	0.021	0.036			

Why it matters. These are the paper's headline econometric results. Market Makers adapted to stress in a way consistent with endogenous-intermediation theories; HFTs mostly continued to trade in their usual short-horizon style.

5.6 Figure 4: HFTs trade ahead of short-run price moves

Read:

- When HFTs are net buyers, prices rise in the next second and remain elevated for up to 20 seconds.
- The effect is strongest for *aggressive* HFT trades and weak or reversed for passive HFT buys.
- The price movement occurs *after* the HFT trade, not during the same second.

Why it matters. This is hard to reconcile with the view that HFTs are merely taking the other side of imbalances and then passively warehousing inventory. It is much easier to reconcile with faster traders getting in front of short-run price changes.

5.7 Table V: direct evidence on quote sniping

Read:

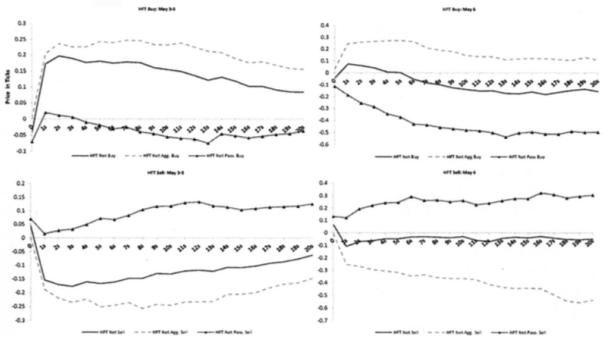

Figure 4. High-Frequency Traders' trading and prices. This figure illustrates how prices change after HFT trading activity in a given second. The upper left panel presents results for buy trades for May 3 through 5, 2010, the upper right panel presents results for buy trades on May 6, 2010, and the lower left and lower right panels present corresponding results for sell trades. For an "event-second" in which High-Frequency Traders are net buyers, net Aggressive Buyers, and net Passive Buyers, value-weighted average prices paid by the High-Frequency Traders in that second are subtracted from the value-weighted average prices for all trades in the same second and each of the following 20 seconds. The results are averaged across event-seconds, weighted by the magnitude of High-Frequency Traders' net position change in the event-second. The upper left panel presents results for May 3 through 5, the upper right panel presents results for May 6, and the lower two panels present results for sell trades calculated analogously. Price differences on the vertical axis are scaled so that one unit equals one tick ($12.50 per contract).

Table V

Shares of Passive and Aggressive Trading Volume Around Price Increase and Price Decrease Events

This table presents each trader category's share of aggressive and passive trading volume for the last 100 contracts traded before a price increase event or price decrease event and the first 100 contracts traded at the new higher (lower) price after a price increase (decrease) event. For comparison purposes, this table also presents the unconditional share of aggressive and passive trading volume of each trader category. Trading categories are High-Frequency Traders (Hft), Market Makers (Mm), Fundamental Buyers (Buyer), Fundamental Sellers (Seller), Opportunistic Traders (Opp), and Small Traders (Small). To emphasize the symmetry between buying and selling, the rows for Buyer and Seller in Panels B and D have been reversed relative to Panels A and C.

Panel A: Trading at the Best Ask Around Price Increase Events, May 3–5, 2010

	Last 100 Contracts		First 100 Contracts		Volume at Best Ask	
	Passive	Aggressive	Passive	Aggressive	Passive	Aggressive
Hft	28.72%	57.70%	37.93%	14.84%	34.33%	34.04%
Mm	15.80%	8.78%	19.58%	7.04%	13.48%	7.27%
Buyer	6.70%	11.61%	4.38%	26.17%	4.57%	21.53%
Seller	16.00%	2.65%	11.82%	7.09%	16.29%	5.50%
Opp	32.27%	19.21%	25.95%	43.39%	30.90%	31.08%
Small	0.51%	0.04%	0.34%	1.46%	0.44%	0.58%

Panel B: Trading at the Best Bid Around Price Decrease Events, May 3–5, 2010

	Last 100 Contracts		First 100 Contracts		All Volume at Best Bid	
	Passive	Aggressive	Passive	Aggressive	Passive	Aggressive
Hft	27.41%	55.20%	38.31%	15.04%	34.45%	34.17%
Mm	15.49%	8.57%	20.64%	6.58%	13.79%	7.45%
Seller	5.88%	11.96%	3.83%	24.87%	5.67%	20.91%
Buyer	17.98%	3.22%	12.71%	8.78%	15.40%	6.00%
Opp	32.77%	20.99%	24.18%	43.41%	30.30%	30.89%
Small	0.47%	0.06%	0.34%	1.32%	0.39%	0.58%

(*Continued*)

Table V—*Continued*

Panel C: Trading at the Best Ask Around Price Increase Events, May 6, 2010

	Last 100 Contracts		First 100 Contracts		All Volume at Best Ask	
	Passive	Aggressive	Passive	Aggressive	Passive	Aggressive
Hft	28.46%	38.86%	30.55%	14.84%	30.94%	26.98%
Mm	12.95%	5.50%	13.88%	5.45%	12.26%	5.82%
Buyer	6.31%	17.49%	5.19%	21.76%	5.45%	20.12%
Seller	13.84%	3.84%	14.30%	5.71%	14.34%	4.40%
OPP	38.26%	34.26%	35.94%	51.87%	36.86%	42.37%
Small	0.19%	0.06%	0.16%	0.37%	0.16%	0.31%

Panel D: Trading at the Best Bid Around Price Decrease Events, May 6, 2010

	Last 100 Contracts		First 100 Contracts		All Volume at Best Bid	
	Passive	Aggressive	Passive	Aggressive	Passive	Aggressive
Hft	28.38%	38.67%	30.13%	14.59%	30.09%	26.29%
Mm	12.27%	5.04%	14.85%	5.64%	12.05%	5.88%
Seller	4.19%	16.46%	3.77%	21.21%	3.82%	17.55%
Buyer	15.83%	5.90%	13.89%	6.97%	15.27%	7.26%
Opp	39.12%	33.86%	37.15%	51.10%	38.56%	42.68%
Small	0.21%	0.08%	0.21%	0.48%	0.21%	0.34%

- Before price increases on May 3–5, HFTs account for 57.70% of aggressive volume at the old best ask, versus 34.04% unconditionally.
- After the ask moves up, their aggressive share at the new ask drops to 14.84%, while their passive share rises to 37.93%.
- Before price decreases, HFTs account for 55.20% of aggressive sell volume at the old best bid, versus 34.17% unconditionally.
- After the bid moves down, their aggressive share falls to 15.04%, while their passive share rises to 38.31%.
- Market Makers are much more passive: for example, before price increases they account for only 7.27% of aggressive volume at the old ask, rising only modestly to 8.78%, but their passive share at the old ask rises from 13.48% to 15.80%.

Why it matters. This is the cleanest microstructure evidence in the paper. HFTs appear to lift stale depth just before the quote moves and then repost at the new quote. Market Makers, by contrast, look like slower passive liquidity suppliers.

6 Short summary

- **Method.** Use account-level E-mini transaction data, classify traders from inventory and volume behavior, and compare second-by-second trading patterns before and during the Flash Crash.
- **Main identification idea.** Keep trader labels fixed from pre-crash days, then test how inventory-price comovement and event-time trading behavior differ by trader type.
- **Main result on inventory capacity.** Intermediaries absorbed only a small fraction of the order imbalance relative to the 75,000-contract sell program.
- **Main result on HFT behavior.** HFTs did not significantly change their inventory dynamics during the price decline, and their trades precede short-run price changes.
- **Main result on Market Makers.** Market Makers exhibit more traditional dealer behavior and change their lag structure during the crash.
- **Bottom line.** The most active fast traders in this market were not acting like textbook stabilizing market makers during the Flash Crash.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the most active nondesignated intraday intermediaries in the E-mini – the HFTs – did not materially change their basic trading pattern during the Flash Crash.
- It also establishes that these HFTs had limited inventory capacity and therefore could not absorb the full liquidity shock.
- Finally, it shows that HFT behavior is more consistent with fast trading and quote sniping than with traditional market making.

7.2 Economic intuition

- A classic dealer stabilizes prices by absorbing inventory when others urgently want to trade.
- A fast trader who profits from speed does something different: trade ahead of short-run price changes, hit stale quotes, then recycle inventory quickly.
- During the Flash Crash, the second behavior seems to describe HFTs better than the first.
- Meanwhile, genuine inventory-bearing intermediaries were too small to absorb the shock, so prices had to adjust sharply.

7.3 Takeaway

This paper is important because it sharpens the meaning of "liquidity provision" in electronic markets. A trader can submit huge volume, trade almost continuously, and still not function like a stabilizing dealer when the market is stressed. The Flash Crash, in this reading, was not a story of HFTs suddenly disappearing; it was a story in which the main fast traders kept doing what fast traders do, while the true inventory-bearing capacity of the market proved too small for the shock.